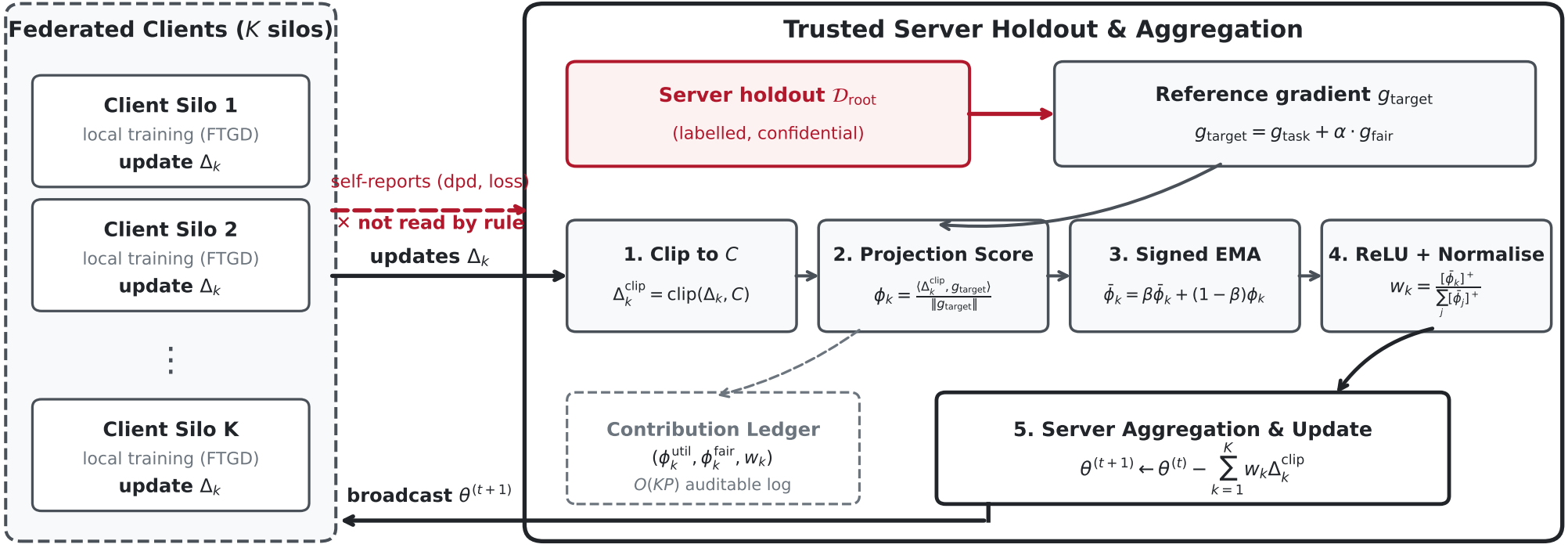
Federated Clients (K silos)
Client Silo 1
local training (FTGD)
update Δ_k
Client Silo 2
local training (FTGD)
update Δ_k
Client Silo K
local training (FTGD)
update Δ_k
self-reports (dpd, loss)
× not read by rule
updates Δ_k
broadcast $\theta^{(t+1)}$
Trusted Server Holdout & Aggregation
Server holdout $\mathcal{D}_{\text{root}}$
(labelled, confidential)
Reference gradient g_{target}
$g_{\text{target}} = g_{\text{task}} + \alpha \cdot g_{\text{fair}}$
1. Clip to C
$\Delta_k^{\text{clip}} = \text{clip}(\Delta_k, C)$
2. Projection Score
$\phi_k = \frac{\langle \Delta_k^{\text{clip}}, g_{\text{target}} \rangle}{\|g_{\text{target}}\|}$
3. Signed EMA
$\bar{\phi}_k = \beta\bar{\phi}_k + (1-\beta)\phi_k$
4. ReLU + Normalise
$w_k = \frac{[\bar{\phi}_k]^+}{\sum_j [\bar{\phi}_j]^+}$
Contribution Ledger
$(\phi_k^{\text{util}}, \phi_k^{\text{fair}}, w_k)$
$O(KP)$ auditable log
5. Server Aggregation & Update
$\theta^{(t+1)} \leftarrow \theta^{(t)} - \sum_{k=1}^{K} w_k \Delta_k^{\text{clip}}$